

Inorganic Chemistry

Science

May, 2026



**SE
TU**

Ollscoil
Teicneolaíochta
an Oirdheiscirt

South East
Technological
University

Short Title: Inorganic Chemistry
Faculty Science and Computing
Department Science
Cluster Chemistry
Author EMLANDERS
Credits 5

Level: Introductory

Description of Module / Aims

This module gives an appreciation of the various bonding theories and structural characteristics of inorganic compounds, and introduces the concept of molecular orbitals. Other topics include dual nature of the electron, chemistry of non-aqueous systems, non-metals including hydrogen, nitrogen, oxygen, the halogens, and metals including the alkali, alkaline and transition metals. An introduction to the role of inorganic chemistry in the field of polymer science and nanotechnology is also provided.

Programmes

CHEM-0028	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	2 / 4 / M
CHEM-0028	BSc in Pharmaceutical Science (WD_SPHSC_D)	2 / 4 / M

Indicative Content

- Electron configuration and periodicity, wave mechanics and Schrodinger's equation, radial distribution functions
- Atomic properties, effective nuclear charge, ionisation energy, electronegativity
- Ionic bonding, ionic crystals, radius ratio, unit cell, lattice energy, typical crystal structures
- Further treatment of covalent bonding, atomic orbitals, molecular orbitals, valence shell electron pair repulsion theory and molecular shape
- Metallic bonding, metallic crystals, band theory
- Intermediate bond types, polar covalent, hydrogen bonding, van der Waals forces
- Non-aqueous systems, protic and aprotic solvents, hard-soft acid-base principle, acid-base titrations in non-aqueous solvents
- An introduction to inorganic polymers, their synthesis and applications
- Inorganic compounds in nanotechnology and their applications
- Chemistry of the non-metals and their compounds: hydrogen, nitrogen group, oxygen group, the halogens and including oxides and hydrides of these elements
- Physical and chemical properties of the alkali, the alkaline and transition metals
- Associated practicals that reflect the content of the module

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Determine electronic structure, coordination number, unit cell type and crystal structure.
2. Compare the wave-particle nature of the electron, radial distribution diagrams, the different solvent systems and the various bonding theories of this module.
3. Calculate lattice energy and effective nuclear charge.
4. Examine the role of inorganic compounds/polymers in nanotechnology including their synthesis/application and the physical and chemical properties of the alkali, alkaline, transition metals and their complexes.
5. Interpret the chemistry of non-metallic elements and their compounds.
6. Complete the preparation of co-ordination complexes in the laboratory.
7. Apply error calculations to volumetric analysis.

Learning and Teaching Methods

- Practicals.
- Lectures.
- Self-directed study.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5
Continuous Assessment	50%	
<i>Practical</i>	50%	6,7

Assessment Criteria

<40%: No understanding of the basic principles of inorganic chemistry.

40%-49%: Has a superficial knowledge of the subject area.

50%-59%: Shows a reasonable level of understanding of the various principles involved.

60%-69%: Shows a good knowledge of the area. Shows ability to reason and analyse problems in this area.

70%-100%: Excellent knowledge of the subject area. Excellent reasoning skills and focused well thought-out arguments.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	36	
Independent Learning	75	

Essential Material

- Browne, T.H. and H.E. LeMay. *Chemistry, The Central Science*. 3rd ed. Australia: Pearson, 2013.
- Cotton, A.F. and G. Wilkinson. *Basic Inorganic Chemistry*. 3rd ed. New York: John Wiley & Sons, Inc, 1995.

Supplementary Material

- De Jaeger, J. and M. Gleria. *Silicon-based Inorganic Polymers*. US: Nova Science Publishers Inc, October 2008.
- Housecroft, C. E. and A.G. Sharpe. *Inorganic Chemistry*. 4th ed. UK: Prentice Hall, 2012.
- Mark, J.E., H.R. Allcock and R. West. *Inorganic Polymers*. 2nd ed. UK: Oxford University Press, 2005.
- Mattoussi, H. and J. Cheon. *Inorganic Nanoprobes for Biological Sensing and Imaging*. US: Artech House, 2008.
- Miessler, G.L. and D.A. Tarr . *Inorganic Chemistry*. 5th ed. UK: Pearson., 2014.
- Rayner-Canham, G. and T. Overton. *Descriptive Inorganic Chemistry*. 6th ed. New York: Macmillan, 2014.

Requested Resources

- Room Type: Chemistry Lab